

Part 1: S5

Use the simplified rules for **S5** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S5**.

1. $\Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$ in **S5**

1.	$\mathbb{F} 1, \Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$ ✓	Assumption
2.	$\mathbb{F} 1, \Box(\Box p \rightarrow (q \vee r))$ ✓	$\vee\mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(\Box q \rightarrow p)$ ✓	$\vee\mathbb{F} 1$
4.	$\mathbb{F} 2, \Box p \rightarrow (q \vee r)$ ✓	$\Box\mathbb{F} 2$
5.	$\mathbb{T} 2, \Box p$	$\rightarrow\mathbb{F} 4$
6.	$\mathbb{F} 2, q \vee r$ ✓	$\rightarrow\mathbb{F} 4$
7.	$\mathbb{F} 2, q$	$\vee\mathbb{F} 6$
8.	$\mathbb{F} 2, r$	$\vee\mathbb{F} 6$
9.	$\mathbb{F} 3, \Box q \rightarrow p$	$\Box\mathbb{F} 3$
10.	$\mathbb{T} 3, \Box q$	$\rightarrow\mathbb{F} 9$
11.	$\mathbb{F} 3, p$	$\rightarrow\mathbb{F} 9$
12.	$\mathbb{T} 3, p$	$\Box\mathbb{T} 5$
	⊗	

The only branch closes, so this is **valid** in **S5**.

$\Diamond\Box(p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q)$ in **S5**

1.	$\mathbb{F} 1, \Diamond\Box(p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond\Box(p \rightarrow q)$ ✓	$\rightarrow\mathbb{F} 1$
3.	$\mathbb{F} 1, \Box p \rightarrow \Diamond q$ ✓	$\rightarrow\mathbb{F} 1$
4.	$\mathbb{T} 1, \Box p$	$\rightarrow\mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond q$	$\rightarrow\mathbb{F} 3$
6.	$\mathbb{T} 2, \Box(p \rightarrow q)$	$\Diamond\mathbb{T} 2$
7.	$\mathbb{T} 2, p \rightarrow q$ ✓	$\Box\mathbb{T} 2$
8.	$\mathbb{T} 2, p$	$\Box\mathbb{T} 4$
9.	$\mathbb{F} 2, q$	$\Diamond\mathbb{F} 5$
$\swarrow \quad \searrow$		
10.	$\mathbb{F} 2, p \quad \mathbb{T} 2, q$	$\rightarrow\mathbb{T} 7$
	⊗ ⊗	

Both branches close, so this is **valid** in **S5**.

$\Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$ in **S5**

1.	$\mathbb{F} 1, \Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$ ✓	Assumption
2.	$\mathbb{T} 1, \Box(p \vee q)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond(\neg q \vee r)$ ✓	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond(p \vee r)$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 2, \neg q \vee r$ ✓	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 2, p \vee q$ ✓	$\Box \mathbb{T} 1$
8.	$\mathbb{F} 2, p \vee r$ ✓	$\Diamond \mathbb{F} 5$
9.	$\mathbb{F} 2, p$	$\vee \mathbb{F} 8$
10.	$\mathbb{F} 2, r$	$\vee \mathbb{F} 8$
└──┬──		
11.	$\mathbb{T} 2, p$ $\mathbb{T} 2, q$	$\vee \mathbb{T} 7$
	⊗	
	└──┬──	
12.	$\mathbb{T} 2, \neg q$ ✓	$\vee \mathbb{T} 6$
13.	$\mathbb{F} 2, q$ ⊗	$\neg \mathbb{T} 12$
	⊗	

All branches close, so this is **valid** in **S5**.

$(\neg \Diamond p \vee \neg \Diamond \Diamond p) \rightarrow \neg \Diamond \Diamond \Diamond p$ in **S5**

1.	$\mathbb{F} 1, (\neg \Diamond p \vee \neg \Diamond \Diamond p) \rightarrow \neg \Diamond \Diamond \Diamond p$ ✓	Assumption
2.	$\mathbb{T} 1, \neg \Diamond p \vee \neg \Diamond \Diamond p$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \neg \Diamond \Diamond \Diamond p$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond \Diamond \Diamond p$ ✓	$\neg \mathbb{F} 3$
5.	$\mathbb{T} 2, \Diamond \Diamond p$ ✓	$\Diamond \mathbb{T} 4$
6.	$\mathbb{T} 3, \Diamond p$ ✓	$\Diamond \mathbb{T} 5$
7.	$\mathbb{T} 4, p$	$\Diamond \mathbb{T} 6$
└──┬──		
8.	$\mathbb{T} 1, \neg \Diamond p$ ✓	$\vee \mathbb{T} 2$
9.	$\mathbb{F} 1, \Diamond p$	$\neg \mathbb{T} 8$
10.	$\mathbb{F} 4, p$	$\Diamond \mathbb{F} 9$
11.	⊗	$\neg \mathbb{T} 8$
12.	$\mathbb{F} 1, \Diamond \Diamond p$	$\Diamond \mathbb{F} 11$
	$\mathbb{F} 3, \Diamond p$	
	⊗	

Both branches close, so this is **valid** in **S5**. There are many ways to close this one; this is the fastest I found, but it's definitely not the only way.

$(\Diamond p \wedge \Box q) \rightarrow \Diamond q$ in **K**

1.	$\mathbb{F} 1, (\Diamond p \wedge \Box q) \rightarrow \Diamond q$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond p \wedge \Box q$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond q$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond p$ ✓	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Box q$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, p$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{F} 1.1, q$	$\Diamond \mathbb{F} 3$
8.	$\mathbb{T} 1.1, q$	$\Box \mathbb{T} 5$
	⊗	

The trunk closes, so it is **valid** in **K**. What's odd about this one is that even though the right hand side of the conditional is just about q , the $\Diamond p$ plays a crucial role in the tableau, by making us introduce $w_{1.1}$.

$\Box(p \wedge \neg p) \vee \Diamond(q \vee \neg q)$ in **K**

1.	$\mathbb{F} 1, \Box(p \wedge \neg p) \vee \Diamond(q \vee \neg q)$ ✓	Assumption
2.	$\mathbb{F} 1, \Box(p \wedge \neg p)$ ✓	$\vee \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(q \vee \neg q)$	$\vee \mathbb{F} 1$
4.	$\mathbb{F} 1.1, p \wedge \neg p$	$\Box \mathbb{F} 2$
5.	$\mathbb{F} 1.1, q \vee \neg q$ ✓	$\Diamond \mathbb{F} 3$
6.	$\mathbb{F} 1.1, q$	$\wedge \mathbb{F} 5$
7.	$\mathbb{F} 1.1, \neg q$ ✓	$\vee \mathbb{F} 5$
8.	$\mathbb{T} 1.1, q$	$\neg \mathbb{F} 7$
	⊗	

This is also valid, and it's also wrong for the same reason. (Not at all on the test digression. This example shows that **K** has a very weird property. It is not Halldén-complete. A logician, Sören Halldén, in the 1950s noticed that some logics have the odd property that $X \vee Y$ is a theorem even though (a) neither X nor Y is, and (b) X and Y have no variables in common. We've come to call such logics Halldén-incomplete, and the opposite property Halldén-completeness. In general, the logics that are interesting and get used in philosophical applications are Halldén-complete.)

$\Box \neg(p \vee \neg q) \rightarrow \Box p$ in **K**

1.	$\mathbb{F} 1, \Box \neg(p \vee \neg q) \rightarrow \Box p$ ✓	Assumption
2.	$\mathbb{T} 1, \Box \neg(p \vee \neg q)$	$\rightarrow \mathbb{T} 1$
3.	$\mathbb{F} 1, \Box p$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1.1, p$	$\Box \mathbb{F} 3$
5.	$\mathbb{T} 1.1, \neg(p \vee \neg q)$ ✓	$\Box \mathbb{T} 2$
6.	$\mathbb{F} 1.1, p \vee \neg q$ ✓	$\neg \mathbb{T} 5$
7.	$\mathbb{F} 1.1, p$	$\vee \mathbb{F} 6$
8.	$\mathbb{F} 1.1, \neg q$ ✓	$\vee \mathbb{F} 6$
9.	$\mathbb{T} 1.1, q$	$\neg \mathbb{F} 8$

The tree is open, so this is not a theorem, it is **invalid** in **K**. The countermodel has two worlds: w_1 and $w_{1.1}$, with $w_1 R w_{1.1}$. At $w_{1.1}$, p is false and q is true.

$\Box(p \rightarrow (q \wedge r)) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond(q \vee s))$ in **K**

1.	$\mathbb{F} 1, \Box(p \rightarrow (q \wedge r)) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond(q \vee s))$ ✓	Assumption
2.	$\mathbb{T} 1, \Box(p \rightarrow (q \wedge r))$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond\Diamond p \rightarrow \Diamond(q \vee s)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond\Diamond p$ ✓	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond(q \vee s)$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 1.1, \Diamond p$ ✓	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.1, \Box(p \rightarrow (q \wedge r))$	$4 \Box 2$
8.	$\mathbb{F} 1.1, \Diamond(q \vee s)$	$4 \Diamond 5$
9.	$\mathbb{T} 1.1.1, p$	$\Diamond \mathbb{T} 6$
10.	$\mathbb{T} 1.1.1, p \rightarrow (q \wedge r)$	$\Box \mathbb{T} 7$
└───┬───┘		
11.	$\mathbb{F} 1.1.1, p$	$\rightarrow \mathbb{T} 10$
12.	$\otimes$	
	$\mathbb{T} 1.1.1, q \wedge r$ ✓	
13.	$\mathbb{T} 1.1.1, q$	$\wedge \mathbb{T} 11$
14.	$\mathbb{T} 1.1.1, r$	$\wedge \mathbb{T} 11$
15.	$\mathbb{F} 1.1.1, q \vee s$	$\Diamond \mathbb{F} 8$
	$\mathbb{F} 1.1.1, q$	$\vee \mathbb{F} 14$
	$\otimes$	

All the branches close, so this is valid.

$(\Diamond\Box p \wedge \Diamond\Box q) \rightarrow \Diamond(p \wedge q)$ in **K**

1.	$\mathbb{F} 1, (p \wedge \Diamond\Box p) \rightarrow \Box p$ ✓	Assumption
2.	$\mathbb{T} 1, p \wedge \Diamond\Box p$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box p$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, p$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond\Box p$ ✓	$\wedge \mathbb{T} 2$
6.	$\mathbb{F} 1.1, p$	$\Box \mathbb{F} 3$
7.	$\mathbb{T} 1.2, \Box p$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{T} 1.2, p$	$\mathbb{T} \Box 7$

This is open, so it isn't valid. The countermodel has three worlds, w_1 , $w_{1.1}$ and $w_{1.2}$. w_1 can access everything, the other two can only access themselves. And p is true at w_1 and $w_{1.2}$, but false at $w_{1.1}$.

$\Box\Diamond\Box\Diamond p \rightarrow \Box\Diamond p$ in **K**

1.	$\mathbb{F} 1, \Box\Diamond\Box\Diamond p \rightarrow \Box\Diamond p$ ✓	Assumption
2.	$\mathbb{T} 1, \Box\Diamond\Box\Diamond p$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond\Box p$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, \Box\Diamond p$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{F} 1.1, \Diamond\Box p$	$4 \Diamond 3$
6.	$\mathbb{T} 1.1, \Diamond p$	$\mathbb{T} \Box 4$
	$\otimes$	

The trunk closes, so this is **valid**. There are a *lot* of ways to do this one; this is one of the more direct.